
Part IV

Concurrent Aerobic and Strength Training Throughout the Lifespan



Concurrent Training in Children and Adolescents

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Martijn Gäbler and Urs Granacher

Introduction

The central theme of this chapter revolves around the additive but also interfering effects occurring in children and adolescents when combining aerobic and strength training (i.e., concurrent training). Because both training status and (biological) age can influence adaptations and training goals, distinctions are made between young athletes and non-athletic youth [1], and between children (boys aged 6–13 years; girls aged 6–11 years) and adolescents (boys aged 14–18 years; girls aged 12–18 years) [2]. Also, the relevance of certain physical fitness components is different depending on age and training goals. Caspersen [3] operationalized physical fitness as consisting of primarily health- and skill-related components (Table 17.1). It has to be noted that the training of health-related components (e.g., muscular strength, endurance) improves athletic performance [4] and that the training of skill-related components (e.g., balance, agility) also produces health benefits (e.g., injury prevention) [5]. As a matter of fact, most skill-related components of physical fitness (reaction time excluded) are considered as health-related fitness components [6]. The distinction made by Caspersen [3] is still useful to highlight the components that are relevant for sports performance.

M. Gäbler · U. Granacher (✉)

Division of Training and Movement Sciences, Faculty of Human Sciences,
University of Potsdam, Potsdam, Germany

e-mail: mgaebler@uni-potsdam.de; urs.granacher@uni-potsdam.de

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Table 17.1 Components of physical fitness categorized as health- and skill-related according to Caspersen [3] and updated according to Bouchard et al. [6]

Health-related fitness components	Skill-related fitness components
Cardiorespiratory fitness (Heart/lung function, blood pressure)	Agility*
Morphological component (BMI, body composition, flexibility)	Balance*
Muscular fitness (Strength, power, endurance)	Coordination*
Motor component (Agility, balance, coordination, speed)	Speed*
Metabolic component (Glucose tolerance, inflammatory markers)	Power*
	Reaction time

Note: Components marked with * are also considered as health-related fitness components

The Importance of Aerobic and Strength Training in Children and Adolescents

It is undisputed that physical activity has positive effects on the health of children and adolescents. The World Health Organization (WHO) [7] recommends children and adolescents aged 5–17 years to accumulate daily at least 60 minutes of physical activity at moderate to vigorous intensity. Most of the daily physical activity should be aerobic in nature. Vigorous-intensity activities should be incorporated, including those that strengthen muscle and bone, at least three times per week. Physical activity includes both aerobic and strengthening activities that can either be accomplished through play and games, but also in organized exercise settings such as physical education classes and sport clubs. However, current guidelines [7] may undermine the development of muscular fitness (muscle strength, muscle power, and muscular endurance) that is a prerequisite for energetic and vigorous movement and play [8]. Moreover, an increasing percentage of youth do not meet the WHO physical activity recommendations [9, 10], which was identified as an exercise deficit disorder that can result in childhood obesity and pediatric dynapenia [9, 11–13]. In 2008, only 54% of children at pre-school age adhered to the recommended amount of physical activity [14]. A survey performed by the WHO in 2009/2010 suggested that this percentage shrinks with increasing age from 23% in 11-year-olds to 15% in 15-year-olds [15].

Besides non-athletic children and adolescents, concurrent training is of specific relevance for the sub-population of young athletes. Williams [1] defined a young athlete as “a child or adolescent who is still growing and maturing toward adulthood and who systematically trains (>once per week) and competes (>1-year competition history) in at least one specific sport.” Given that most types of sport require some combination of strength, endurance (and speed) [16], training in young athletes should at least involve a sufficient stimulus to help develop these components of

physical fitness. It is not surprising that adding strength exercises to an existing training program in young athletes aged 6–18 years resulted in improved skill-related components of physical fitness (e.g., speed and power) [4]. Thus, participation in concurrent training can help to meet the WHO requirements for physical activity, promote health, and improve movement skills and athletic performance.

Promoting Physical Development Throughout Childhood and Adolescence

The physical development of children and adolescents needs to be taken into account carefully when administering any form of physical training. Several models [17–20] have been proposed as a guide to systematically develop components of physical fitness, such as strength, endurance, speed, and movement skills throughout childhood and adolescence. The essence of these models is to optimize training for young athletes in order to achieve their optimal physical performance potential in adulthood, thus favoring long-term training goals over short-term success. Opposing the restricted focus on chronological age [21], these models acknowledge that the development of physical fitness components is not a linear process but is rather related to biological development. Therefore, training in youth should address those physical fitness components that show a high sensitivity to training stimuli. Furthermore, despite their focus on the physical development of young athletes, these models can best be viewed as “a pathway for all youth” [18, 20] that promote fun, confidence, and lifelong engagement in physical activity. As such, they are useful guides to individualize training programs for children and adolescents related to their biological age and prior training experience.

The first model to relate biological development to training and trainability of children and adolescents is the long-term athlete development (LTAD) model [17, 18]. This model uses the onset of peak height velocity (PHV) as a reference point. PHV is the age during adolescence at which an individual shows the largest change/growth in stature, and can be estimated from simple anthropometric measures (i.e., sitting and standing height), sex, and chronological age [22]. A free online PHV calculator can be found at the University of Saskatchewan website [23]. The inclusion of PHV assessment in fitness promotion in children and adolescents allows shaping training differently and individually for early, normal, and late maturing children. Thus, the LTAD model uses the onset of PHV as a reference to determine “windows of opportunity” as critical periods in physical development that are crucial for achieving one’s athletic potential [17, 18, 24]. However, a lack of empirical evidence in support of “windows of opportunity” led to the most important criticism on the LTAD model [19, 25].

Out of discontent with the LTAD model [17, 18], Lloyd et al. [19] proposed the youth physical development (YPD) model, which they later modified and titled the composite youth development (CYD) model [20]. The YPD model (later the CYD model) assumes that physical fitness components are trainable throughout childhood and adolescence, but the importance to train different components changes

with maturation. Similar to the LTAD, onset of PHV is used as a reference point to determine biological age. Furthermore, there are different CYD models for males and females and the model should be adapted according to the initial training status [19, 20]. The CYD model [20] further suggests that emphasis in training should be guided by the maturational status according to the years removed from PHV, growth rate, and the related developmental stages of the individual.

According to the CYD model, the importance of developing certain components of physical fitness changes over developmental stages. Early childhood is a period of rapid growth that starts roughly at the chronological age of 2–4 years. During this developmental stage, physical activity should be unstructured, playful, and fun. Developing fundamental movement skills (FMS) and muscle strength should be emphasized [19, 20, 26]. The remaining years pre-PHV involve the developmental stage of middle childhood and are associated with a steady increase in body height. This stage lasts usually longer for males (5–13 years) than for females (5–11 years) [27, 28]. During this period training and physical activities should be highly variable without specialization in a single sport [29]. Thus, emphasis should still be placed on developing muscle strength and FMS, and to a lesser extent on developing mobility, agility, speed, and power [19, 20]. The onset of adolescence is signified by an increase in growth rate that peaks at PHV and slowly declines thereafter until adulthood [27, 30]. Adolescence roughly lasts from the age of 14–20 years for boys and from the age of 12–18 years for girls. Training structure should be gradually increased towards adulthood and should specialize in a specific sport [19, 20, 29]. Emphasis should be placed on developing sport-specific skills, agility, speed, muscle power and strength and, after PHV, also muscle hypertrophy.

Training for endurance and metabolic conditioning become more important towards late adolescence but training goals may differ depending on the population and the practiced sport. Thus, there might be individual differences in the relative importance of certain components of physical fitness. Particularly young endurance athletes may want to expose themselves to larger volumes of aerobic training in order to develop a high aerobic capacity [31]. Taken together, the CYD model suggests that the development of muscle strength and movement competency should have a central position for both children and adolescents while the importance of aerobic training gradually increases over the developmental stages [20].

The central position of muscle strength in the CYD [20] seems contradictory to the views of the WHO [7], who emphasizes aerobic activity. This contradiction can probably best be understood through a distinction in training goals. The main concern of the WHO is to promote health by reducing physical inactivity, overweight, and obesity, while the CYD aims at promoting physical development, movement skill competency, and lifelong sports participation. Furthermore, when strength development is neglected and muscle strength levels are deficient, a negative cascade threatens to manifest itself that has been referred to as an exercise deficit disorder [11]. The cascade starts with reduced motor skill competence, leading to low movement confidence and increased sedentary behavior which ultimately results in adverse health outcomes later in life [28]. Closely related to this exercise deficit is the recent observation of pediatric dynapenia [8, 13]. Dynapenia is a deficiency in muscle strength that is usually found in old adults [32]. Recently, Faigenbaum [13]

described pediatric dynapenia as an acquired condition not caused by neurologic or muscular disease that is reversible with strength- and skill-based activities [8].

Consequently when recommendations are combined, it can be concluded that strength should be developed to promote physical development in youth [19, 20], while a minimum amount of aerobic activity (60 min/day) is recommended for positive health outcomes [7]. In a recently published narrative review paper, Pichardo et al. [33] summarized the available models on LTAD and integrated those into a new conceptualized approach.

Single-Mode Strength Training

Given the crucial effects muscle strength has on different physical fitness components [34], it is not surprising that national and international associations such as the WHO [7], the American Academy of Pediatrics [35], the International Olympic Committee [36], and the National Strength and Conditioning Associations of America [37] and the United Kingdom [21] recommend strength training throughout youth. Strength training can be started as soon as children are capable of understanding and following instructions [38], which usually comes down to a minimum calendar age of 6 years [39]. It should, however, be highlighted that learning the proper exercise technique and supervision by a strength and conditioning professional are necessities for safety purposes.

Effects

Meta-analyses generally show positive effects of strength training in youth on muscular fitness (i.e., muscular power, muscular strength, local muscular endurance [40]) and movement skills (see Table 17.2 for an overview). Payne et al. [41] found moderate effect sizes ($ES = 0.75$) of strength training on measures of muscular strength and muscular endurance in non-athletic youth (≤ 18 years). Isokinetic training programs seemed to be less effective than isometric or isotonic programs. Age and sex were not found to modify the effectiveness of strength training but it should be noted that the effect sizes in this study were not corrected for small sample sizes nor weighted. Behm et al. [42] analyzed the pre to post differences of 107 studies and revealed that strength training had small effects on sprinting ($ES = 0.48$), moderate effects on jumping ($ES = 0.53$), and large effects on lower body strength ($ES = 1.14$) in children and adolescents. Sub-analyses revealed that children and non-athletic youth responded to training with a larger magnitude in effect sizes.

Two more recent meta-analyses [43, 44] found that strength training in non-athletic youth (≤ 18 years) had large effects ($ES = 1.12$) on muscular strength [43], and moderate effects ($ES = 0.52$) on the motor skills running, jumping, and throwing [44]. Furthermore, age was negatively associated ($r = -0.25$) with training-induced improvements of movement skills [44], but adolescents showed larger gains than prepubertal children in muscular strength [43]. This finding was supported by an early meta-analysis [45] that indicated only moderate effects

Table 17.2 Effect sizes of strength training in youth on muscular fitness, movement skills, and sport-specific performance calculated in several meta-analyses

Study	Population	Measure of effect size	Muscular fitness ^a	Movement skills ^b	Sport-specific performance ^c
Payne et al., 1997 [41]	Non-athletic youth ≤18 years	Mean Cohen's d	0.75	–	–
Behringer et al., 2010 [43]	Non-athletic youth ≤18 years	Weighted mean Hedges' g	1.12	–	–
Behringer et al., 2011 [44]	Non-athletic youth ≤18 years	Weighted mean Hedges' g	–	0.52	–
Falk & Tenenbaum, 1996 [45]	Non-athletic youth Girls ≤12 years/ boys ≤13 years	Mean Cohen's d	0.57		
Harries et al., 2012 [46]	Adolescent athletes 13–18 years	Weighted mean difference (cm) ^d		3.08 ^d	
Lesinski et al., 2016 [4]	Young athletes 6–18 years	Weighted mean Hedges' g	1.09	0.58; 0.68; 0.80	0.75
Behm et al., 2017 [42]	All youth <18 years	Within group standardized mean difference	1.14	0.48; 0.53	–

^aMuscular fitness includes measures of muscular strength, muscular power, and muscular endurance

^bMovement skills include sprinting, jumping, and tests of agility

^cSport-specific performances include measures such as throwing and kicking velocity

^dHarries et al., 2012 [46] reported no effect sizes, but absolute differences in jump height

(ES = 0.57) of strength training on measures of muscular strength in children (boys ≤13 years, girls ≤12 years). Two more meta-analyses have been published that specifically focused on young athletes [4, 46]. Harries et al. [46] found large effects of strength training on jump performance (mean difference = 3.1 cm) in adolescent athletes aged 13–18 years. They concluded that plyometric training resulted in the greatest improvements. Moreover, in a recent meta-analysis, Lesinski et al. [4] analyzed studies that included both child and adolescent athletes and found large effects of strength training on muscular strength (ES = 1.09) and vertical jump performance (ES = 0.80). Strength training had moderate positive effects on movement skills such as sprinting (ES = 0.58), agility (ES = 0.68), and sport-specific performance (ES = 0.75). The influence of age on training outcomes was reversed in athletes [4] compared to untrained peers [43, 44]. In athletes, it was found that children, compared to adolescents, showed greater benefits from strength training in terms of muscle strength, but not in terms of movement skills, especially for sport-specific performances. Faigenbaum et al. [28] suggested that dose–response relationships

may differ between trained and untrained children and adolescents (see next section). The greater training history adolescent athletes likely have over child athletes and untrained youth may explain the relative smaller training-induced gains in adolescent athletes. From these data it can be concluded that strength training can improve muscular strength, movement skills, and sports performance in children and adolescents and that age and training status have a moderating effect on these outcomes.

Dose–Response Relationships and Recommendations

Behringer et al. [44] found a positive linear relationship between intensity (% of the one-repetition maximum [1-RM]) and improvements in movement skills (throwing, jumping, and running) of untrained children and adolescents (≤ 18 years). Their data suggested that a minimum intensity of 50% of 1-RM was required to improve movement skills. Strength training parameters in young athletes aged 6–18 years can be derived from Lesinski et al.'s analysis of dose–response relationships between strength training parameters and improved muscular strength [4]. The largest gains in muscular strength were found when training with free-weights, at a frequency of two to three times per week, with five sets per exercise at an intensity corresponding to 80–89% of the 1-RM. However, future studies need to verify safety of training with such high loads in young athletes or untrained youth [28]. Expert opinion [2] on the progression of intensity deems it as a safe practice to learn the proper exercise technique with low loads before progressing to higher loads. Specific recommendations for youth starting with strength training can be found in the review articles of Behm et al. [47] and, more recently, Faigenbaum et al. [34]. According to these guidelines, strength training should be performed on 2 or 3 nonconsecutive days per week, performing 1 or 2 sets of 8–15 repetitions in 8–12 different exercises at an intensity of about 60% of the 1-RM. After establishing proper exercise technique, the child or adolescent can progress to 2–4 sets of 6–12 repetitions at an intensity of up to 80% of the 1-RM [21]. At this stage, more advanced movements could be introduced, [47] such as Olympic weight lifting (See Lloyd et al. [48] for guidelines on the progression towards advanced weight-lifting techniques). As training experience and exercise competence increase, intensity can be progressed even further (<6 repetitions at >85% of the 1-RM [4]). These recommendations are additionally summarized in Table 17.3.

Adaptations

Increased muscle strength can be accomplished both by neural [49–52] and muscular [51–55] adaptations. Training-induced neural adaptations include increased motor unit recruitment and firing rates, increased antagonist muscle inhibition, and improved inter- and intramuscular coordination. The major muscular adaptation leading to increased strength is muscle hypertrophy, but changes in muscle pennation

Table 17.3 Summary of evidence and expert-based recommendations for strength training, aerobic training, and physical activity according to skill level

Parameters	Low skill	Medium skill	High skill
<i>Strength training</i>			
Frequency (w ⁻¹)	2	2–3	2–4
Sets per exercise (N)	1–2	2–4	Multiple
Repetitions per set (N)	8–15	6–12	≤6
Intensity (% of 1-RM)	≤60%	≤80%	≥85%
<i>Aerobic training</i>			
Frequency	3–4 sessions per week		
Duration	40–60 min		
Intensity	85–90% of maximum heart rate		
<i>Physical activity</i>			
Frequency	7 days per week		
Duration	≥60 min		
Intensity	≥50% of maximum heart rate		

Note: Recommendations for physical activity should be viewed as integrated in either strength or aerobic training. E.g., 60 min of high-intensity running counts as both aerobic training and physical activity. Recommendations are based on Behm et al. [47], Faigenbaum et al. [34], Lesinski et al. [4] for strength training; Armstrong et al. [61] for aerobic training; and WHO [7] for physical activity

angle [55], in myosin isoforms [56, 57], as well as in tendon stiffness [58] can also contribute to increased strength. Recently, Legerlotz et al. [58] described the physiological adaptations of young athletes following strength training in their review. They concluded that training-induced improvements in motor skills in children are related to improved neuromuscular control and morphological adaptations of muscle and tendon, although muscle hypertrophy seemed limited. Because prepubertal children display only minimal training-induced muscle hypertrophy [59], it is generally believed that training-induced strength gains in this population reflect neural adaptations. Furthermore, children do not possess the levels of androgens required for muscle hypertrophy [19, 58]. It was indeed found [60] that children aged 10 years showed training-induced gains in elbow flexor strength that were accompanied by increased EMG, but not by increased upper arm circumference. In contrast to children, adolescents did demonstrate muscle hypertrophy following mechanical loading [58], which, together with neural adaptations, may have contributed to increased strength. It is likely to assume that this is related to the changes in hormonal levels (i.e., increased testosterone) with puberty, which may be crucial for muscle hypertrophy to occur [19, 24, 27]. These differences in adaptations again highlight the relevance of biological age in the planning of training and exercise prescription.

Single-Mode Aerobic Training

Unlike the clear benefits of strength training in youth, the trainability of measures of aerobic fitness appears rather limited [62]. A compound measure of aerobic fitness is the maximum rate of oxygen consumption ($\text{VO}_2 \text{ max}$) that depends on the

maximum cardiac output (stroke volume and heart rate) and the arterial-venous oxygen difference. However, due to the lack of a plateau in the VO_2 curve in many youth when exercising to exhaustion, the peak oxygen uptake (peak VO_2) is the preferred measure of aerobic fitness [63]. VO_2 is usually expressed in relation to body mass (in $\text{mL kg}^{-1} \text{min}^{-1}$). When normalized to body mass in non-athletes, peak VO_2 remains quite constant in males during growth and maturation, whereas females may even show a decline [31].

Payne and Morrow [64] found that aerobic training had small to large effects on VO_2 max in children but the magnitude depended on the study design (longitudinal: $ES = 0.35$, cross-sectional: $ES = 0.94$). Another systematic review [65] revealed that a 5–6% increase in peak VO_2 was observed following aerobic training in youth aged 5–16 years. Furthermore, some cross-sectional studies provide indirect evidence for the trainability of peak VO_2 in young individuals, because young athletes demonstrated higher peak VO_2 values than their untrained peers [31].

The observed differences in peak VO_2 values are generally attributed to differences in stroke volume, suggesting that any attempts at improving aerobic capacity in youth should aim at improving stroke volume. In addition, training intensity appears to be a crucial factor in improving peak VO_2 . Narrative and systematic reviews [31, 65, 66] demonstrated that only training at intensities close to the maximal heart rate (85–90%) but not at lower intensities (<80% of maximal heart rate) had an effect on peak VO_2 in youth. Based on these findings, Armstrong et al. [61] suggested that aerobic training in children and adolescents should be performed three to four times per week, with a duration of 40–60 min at an intensity corresponding to 85–90% of maximum heart rate (see also Table 17.3).

Concurrent Aerobic and Strength Training in Healthy Children and Adolescents

Single-mode aerobic and strength training programs may be used to improve components of physical fitness and have benefits for health and sports performance. However, the principle of training specificity [51] dictates that the range of adaptations following single-mode training is limited. Thus, concurrent aerobic and strength training may be preferred to improve a broader range of physical fitness components simultaneously. Interestingly, the result of combining both training modes in adults does not equal the sum of adaptations to single-mode strength and aerobic training. Especially adaptations to strength training (e.g., muscular fitness and muscle hypertrophy) can be compromised when aerobic exercise is performed concurrently compared to single-mode strength training [67]. The potential interfering effects of combining aerobic and strength training modes are discussed extensively in Part II of this book. Of relevance for this chapter is that a high volume of aerobic exercise may limit strength development by interfering with muscle hypertrophy, due to conflicting muscular adaptations [68]. However, there are also data [69] suggesting that interference may be more pronounced in maximum muscle strength and power (i.e., neuromuscular adaptations) rather than in muscle

hypertrophy. It should be noted that our current knowledge is almost exclusively drawn from studies in the adult population [67]. Because of scant literature, much less is known about concurrent training in children and adolescents. Importantly, due to growth and maturation, findings from studies in adults do not simply translate into youth populations [70, 71]. The adaptations to strength training described in a previous section of this chapter emphasize the importance of acknowledging biological age when designing exercise interventions. Because prepubescent children show little capacity for muscle hypertrophy, it could be speculated that there is only minimal interference of aerobic exercise on strength training adaptations. However, should interference occur in neural adaptations [69], it is relevant to take heed when planning training programs for children. Adolescents, on the other hand, show training-induced muscle hypertrophy next to neural adaptations and interference mechanisms may be at work when aerobic and strength training are combined.

Until recently, the only reviews that have been published on concurrent training in children and adolescents have been performed with studies in overweight individuals [72, 73] and will be discussed in a later section of this chapter. More recently, we have performed a systematic review and meta-analysis on the effects of concurrent training in healthy children and adolescents, with a special focus on young athletes [74]. Studies on concurrent training traditionally compare one group that performs both aerobic and strength training to a group that performs only single-mode training. Our meta-analysis [74] involved 15 studies, of which 11 were conducted in endurance athletes (runners, swimmers, rowers, and cross-country skiers) and 4 studies involved non-athletic children and adolescents. The studies involving athletes allowed comparisons between aerobic training and concurrent training, while the studies in untrained children and adolescents allowed only comparisons between strength training and concurrent training. The main results of the meta-analysis are presented in Fig. 17.1.

Additional Strength Training to Enhance Aerobic Performance

The addition of strength training to an aerobically dominant exercise protocol had small effects on time trial performance ($ES = 0.41$) in young athletes aged 10–18 years [73]. A sub-analysis revealed that effect sizes for adolescents were moderate ($ES = 0.52$) and only trivial for children ($ES = 0.17$). These findings suggest that aerobic training combined with strength training improves running, swimming, rowing, and cross-country skiing performance more than single-mode aerobic training in young athletes, especially in adolescents. Further data in the included studies indicated that concurrent training had trivial positive effects ($ES = 0.04$) on aerobic capacity ($\dot{V}O_2$ max or peak $\dot{V}O_2$), and exercise economy ($ES = 0.16$) ($\dot{V}O_2$ at a given exercise intensity). These findings suggest that concurrent training can improve time trial performance in young athletes more than aerobic training alone. However, aerobic capacity and exercise economy appear unaffected by additional strength training. It should be noted that the majority of the included studies did not assess aerobic capacity and exercise economy. Further research is needed to elucidate the underlying mechanisms that contributed to improved athletic performance.

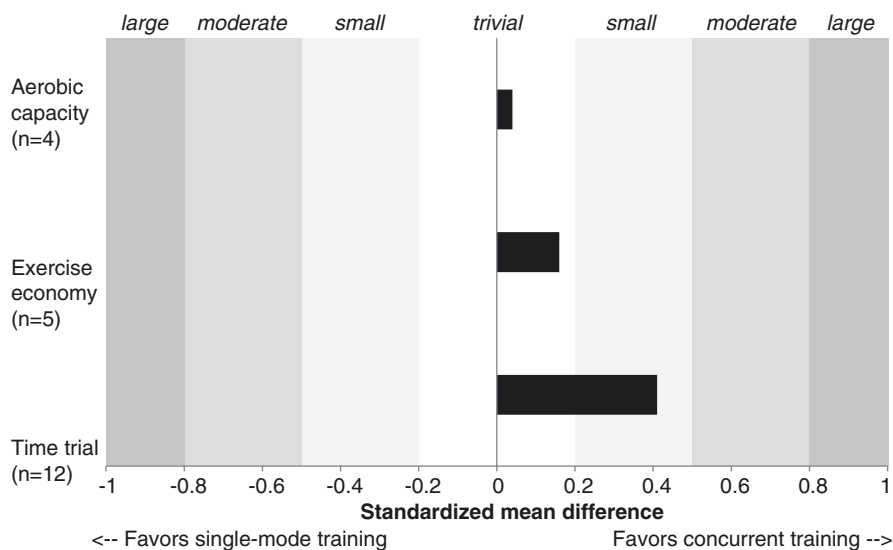


Fig. 17.1 Effects of concurrent training versus singular aerobic training on different outcome parameters in young athletes aged 10–18 years. Positive effect sizes are in favor of concurrent training. N represents the number of training groups (intervention arms) included in the meta-analysis [74]. Bars represent weighted standardized mean differences in aerobic capacity (peak VO_2 or $\text{VO}_{2\text{ max}}$), exercise economy (VO_2 at a given exercise intensity), and athletic performance (time trial)

Probably the most sophisticated study on this topic was performed by Mikkola et al. [75] in a group of 25 young distance runners with a mean age of 17 years. Instead of adding extra training hours like most other studies do, they replaced 19% of the aerobic training volume by explosive type strength training, three times a week for 8 weeks. This procedure ensured that training volume was similar between both groups (aerobic training: 8.5 h/week; concurrent group: 8.8 h/week). Strength exercises performed included sprints (30–150 m), various jumps and weight-bearing exercises, in an attempt to strengthen the major muscle groups of the lower limb and core. The weight-bearing exercises were performed in 2–3 sets of 6–10 repetitions with low loads performed at high movement velocity. The aerobic training consisted mainly of running below the individually determined anaerobic threshold. The authors found that isometric leg-press strength (8% vs. 3%) was increased to larger magnitude in the concurrent compared to the aerobic training group and that these improvements were associated with increases in neural activity ($r = 0.7$) of the knee extensor muscles, while also thickness of the quadriceps femoris muscle increased (4% vs. 2%). What was more interesting for this specific population was that the training had an effect on running economy, without affecting $\text{VO}_{2\text{ max}}$. The oxygen consumption at submaximal speeds was reduced following concurrent training (–3% at 14 km/h), but not following singular aerobic training (+3% at 14 km/h). These results suggested that explosive strength training reduced the energy cost of running in trained adolescent runners. To speculate on the underlying mechanisms, the

improved running economy might be the result of increased neuromuscular activity of type I motor units [76]. Possible training-induced adaptations in muscular and/or tendinous stiffness led to increased muscle and tendon contractility, which allowed for an increased storage of elastic energy in the elastic components of the musculo-tendinous system [76]. As a result, less energy is lost during each gait cycle, which means that less energy needs to be produced to maintain running speed.

No Interference of Additional Aerobic Training on Strength Development

To study the interference effect of aerobic training on strength, power, and muscle hypertrophy, a design where the effects of concurrent training are compared with single-mode strength training should be used. However, our systematic review identified only a few studies that adopted such a design in untrained children [77–79] and adolescents [80]. These identified studies were from the same research group and contained an experimental series with similar research methods. All studies were conducted at the same school cluster in Portugal and comprised one or more concurrent training groups, a single-mode strength training group and a passive control group. The strength training program was performed twice weekly for 8 weeks and consisted of medicine ball throws, jumping, and sprinting exercises. Additionally, also twice weekly, the concurrent training groups performed a sub-maximal 20-m shuttle run test as aerobic training. The aggregated data [74] of these studies showed that the addition of aerobic training to a strength training program had a small effect on leg power, as indicated by vertical jump height ($ES = 0.23$). The low heterogeneity in outcomes across training groups suggested that training adaptations in children and adolescents were similar. These findings imply that there was not an interfering, but an additive effect of aerobic training on power development (vertical jump height) in untrained children and adolescents. However, the methodological limitations of these four studies allow other interpretations as well. A potential confounding factor was the use of the 20-m shuttle run test both as a training and assessment tool. The repeated changes of direction in this test required constant accelerations and decelerations of the entire body. The shuttle run test may have been a sufficient stimulus to increase rapid force production in the lower limbs, which expressed itself in increased vertical jump height. Other reasons for the absence of an interference effect are the untrained status of the participants and the low training volume, which should be kept in mind when interpreting these data.

Sequencing

Under specific circumstances (e.g., high training volume in young athletes), aerobic and strength training have to be combined within one training day. The sequence in which the individual components of aerobic (A) and strength (S) training are performed may affect training adaptations (see Chaps. 10 and 11). So far, two studies are available that examined sequencing effects in youth [81, 82]. These studies

included young soccer players aged 14 (around PHV) and 17 years (post PHV). Here we calculated Hedges' *g* [83] from the mean differences (i.e., post – pre) and pooled standard deviations presented in the original data to determine which sequence (i.e., aerobic training before strength training [A → S] or vice versa [S → A]) produced the largest effects on muscle strength and power, muscle hypertrophy, and aerobic capacity. Figure 17.2 presents the Hedges' *g* and examples for each outcome. Two studies in adult soccer players [84] and non-athletes [85] were also included in Fig. 17.2 for comparative reasons. Although many effect sizes turned out to be trivial in magnitude, some interesting discrepancies between youth and adults were observed which will be discussed in the following.

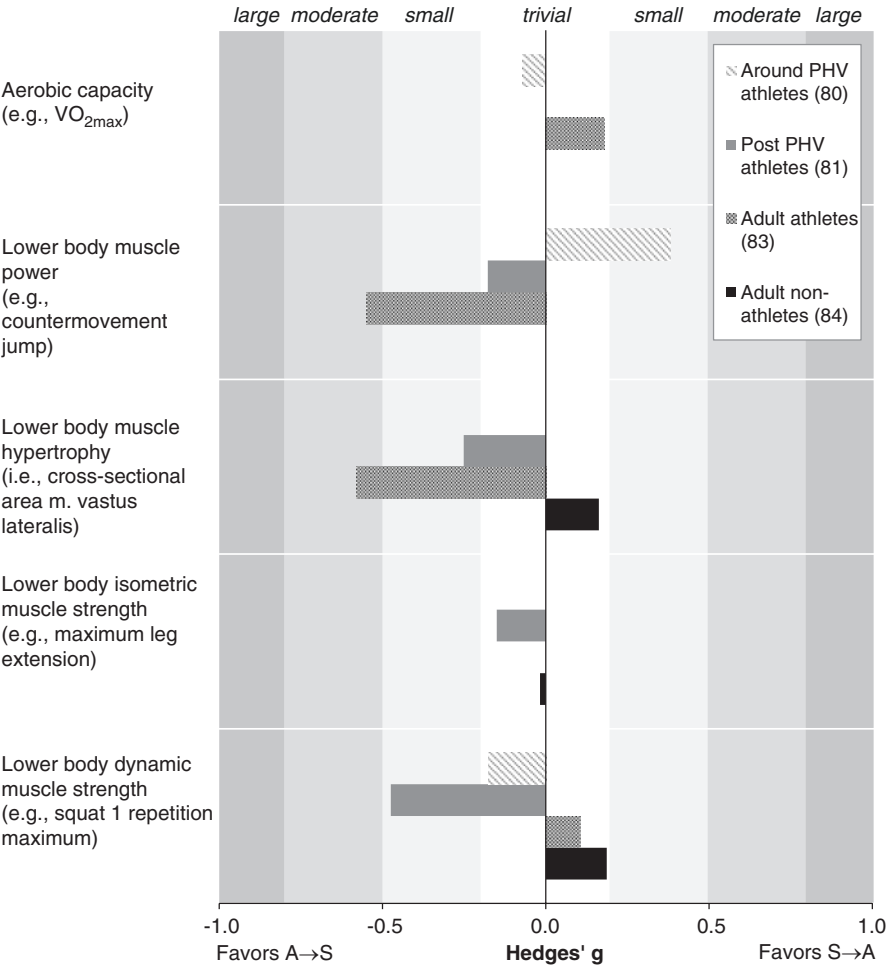


Fig. 17.2 Sequencing effects of aerobic and strength training on aerobic capacity, muscle strength and power, and muscle hypertrophy in different populations. Effect sizes were calculated from the mean differences of the respective groups (A → S and S → A) and their pooled standard deviations [83]

In around PHV soccer players [81], small effects were found for proxies of muscle power in favor of $S \rightarrow A$ and trivial effects were observed for aerobic capacity and dynamic muscle strength in favor of $A \rightarrow S$. Another experimental group in this study [81] performed A and S on alternating days, thereby allowing longer recovery time between training sessions. Makhoul et al. [81] reported that the performance of A and S on one day resulted in similar or greater improvements in aerobic capacity, muscle strength and power compared with the group that performed A and S on alternating days. Another study in post PHV soccer players [82] revealed small effects on measures of dynamic muscle strength and muscle hypertrophy and trivial effects on measures of static muscle strength and proxies of muscle power all in favor of $A \rightarrow S$. The discrepancy in findings between around PHV and post PHV soccer players could be due to differences in maturational status of the examined cohorts. In post PHV youth, the hormonal status allows training-induced muscle hypertrophy [19]. In this context, Baar [86] suggested that $S \rightarrow A$ sequencing may compromise protein synthesis and thus muscle hypertrophy following strength training. This interference could specifically be attributed to post PHV youth.

In adult soccer players, moderate effects were found for proxies of muscle power and muscle hypertrophy in favor of $A \rightarrow S$ and trivial effects were observed for aerobic capacity and dynamic muscle strength in favor of $S \rightarrow A$. For non-athletic adults, trivial effects were detected for muscle hypertrophy as well as measures of static and dynamic muscle strength. A consistent finding in adults appears that $S \rightarrow A$ is favored over $A \rightarrow S$ if the goal is to improve dynamic muscle strength [84, 85]. This was in fact confirmed by two recent meta-analyses [87, 88].

The abovementioned findings indicate a discrepancy between the adult and youth literature on sequencing effects of strength and aerobic training regarding improvements in dynamic muscle strength. While adults primarily benefit from $S \rightarrow A$, youth benefit from $A \rightarrow S$. Children and adolescents compared to adults generally demonstrate greater resistance to fatigue following high-intensity exercise [89]. Strength training quality may therefore be less affected by prior aerobic training when youth perform $A \rightarrow S$ compared to adults. Another important distinction is the difference in training status between the young and adult populations in the aforementioned studies. It was previously suggested that untrained individuals have a greater adaptive reserve regardless of the training mode (A or S) [90]. Taken together, preliminary data suggest that youth may be less susceptible than adults to interfering effects due to better resistance to fatigue. In around PHV soccer players, $S \rightarrow A$ may be favored over $A \rightarrow S$ to improve muscle power. Further, post PHV athletes may benefit most from $A \rightarrow S$, because $S \rightarrow A$ could be more susceptible to interfere with muscle hypertrophy.

Concurrent Training for Health

Given the increasing number of obesity and rising levels of inactivity in young individuals, scientific interest has grown for exercise interventions to promote health already in young populations. To evaluate the effectivity of such interventions,

García-Hermoso et al. [73] have conducted a meta-analysis on data from 12 studies on the effects of concurrent training on body composition and metabolic outcomes in obese and overweight children aged 14–18 years. They specifically included the results of studies that compared a concurrent training group to an aerobic training group and found that additional strength training had small to intermediate effects on body composition (ES: 0.14–0.47), low-density lipoprotein cholesterol (ES: 0.35), and adiponectin concentrations (ES = 0.79), suggesting improved metabolism. These data suggested that concurrent training is more effective than single-mode aerobic training in controlling body weight, fat percentage, and reducing the risk of cardiac and metabolic diseases. However, the authors did not report whether strength training was performed additionally to, or as a replacement of aerobic training, which limits the interpretation of this review. In another meta-analysis, Marson et al. [72] looked specifically at studies that evaluated the effects of different training modes on factors associated with insulin resistance in obese and overweight children and adolescents aged 9–17 years. Increased insulin resistance is associated with obesity and metabolic symptoms later in life. Marson et al. [72] found that aerobic exercise was most successful in decreasing fasted state blood insulin levels, while strength and concurrent training showed only limited effects. Blood glucose levels were changed only minimally following any mode of exercise. The results suggested that aerobic exercise should be part of training in order to decrease resistance to insulin, which may reduce adverse outcomes later in life.

In two of the Portuguese studies mentioned in the previous section [79, 80], body fat was measured in healthy young populations aged 12–14 years. Body fat was reduced the most in concurrent training groups (2.6–2.8%), followed closely by the strength training groups (2.1–2.4%), and to a lesser extent in the passive control groups (0.0–0.9%). These results suggest that body fat can be reduced in young people by performing strength training as well as aerobic training. In studies assessing fat percentage in young athletes [75, 91, 92] both concurrent (–1.6% to +4.0%) and aerobic training (0.2–6.0%) seemed to increase, rather than decrease percentage body fat. Although a similar trend as in non-athletic populations was found, it is arguable that young athletes usually have normal or low levels of body fat and that further reductions are not desired. Taken together, concurrent training has greater and more diverse effects on body composition and other health-related factors than singular strength or singular aerobic training but it remains unclear to what extent this is the result of an increased training volume or of the higher diversity in exercise modes (i.e., aerobic and strength exercises).

Recommendations for Concurrent Training in Youth

The distinction between children and adolescents is important for the design of training programs. Age at PHV can be used as a valid measure to determine the developmental stage of an individual. Unlike in children, strength exercise in adolescents can promote muscle hypertrophy, but careful planning is required. Just as in adults, when training is planned poorly (e.g., aerobic exercise immediately

after strength exercise) and training volume is high, interfering effects of concurrent aerobic and strength training may be expected in adolescents.

Recent models on physical development dictate that training in prepubescent children should focus mainly on developing fundamental movement skills and (neuro-)muscular strength. At this age, strength training should focus on developing the correct exercise technique using loads corresponding to 60% of 1-RM. With increasing maturity, age, and (strength) training experience, loads can gradually progress up to 80% of the 1-RM and more advanced exercises can be introduced. The training of movement skills should become increasingly sport specific. Training with higher loads (>85% of 1-RM) should only be performed when exercise competence is high.

Aerobic activity should be part of any training program throughout all developmental stages due to its positive effects on health-related outcomes. When the goal is to increase aerobic capacity, intensity of training should be sufficiently high (85–90% of maximal heart rate).

The interference effect following concurrent training seems of little concern for untrained and recreationally trained children and adolescents due to overall low training volumes. When aerobic and strength training are performed within a single day or a single training session, aerobic training should not be performed shortly after strength training in young athletes that are post PHV when the goal is to improve muscle strength and hypertrophy. In younger athletes that are around PHV, improvements in leg power (but not strength) may be greater when strength training is performed before aerobic training. These recommendations are based on only two studies and therefore require confirmation. Furthermore, training volume should be gradually increased with increasing age. To reduce overuse injury risk, the total number of training hours per week is recommended not to exceed the number of chronological age and the ratio of training to play time should not exceed 2:1 [29]. Training volume (dose) must be raised when age and training history increase in order to elicit an adaptive effect (response) [28]. From these guidelines, it follows that weekly training volume in children could be lower than in adolescents and thereby regeneration time is increased. This, may in fact help to reduce the risk of interfering training adaptations [90].

Summary

Systematic concurrent aerobic and strength training appears ideal to meet physical activity recommendations, to promote health- and skill-related components of physical fitness and to improve athletic performance in children and adolescents. No interference, but rather an additive effect in training adaptations was observed when youth combine aerobic and strength training compared to single-mode training. Especially adolescents improved their performance in time trials after concurrent training compared to aerobic training alone. However, preliminary evidence suggested that aerobic training should not be performed in short sequence to strength training in adolescents post PHV. Finally, concurrent aerobic and

strength training can help with weight management and reduce the risk of acute and chronic adverse outcomes in young people with inactive lifestyles, overweight, or obesity.

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